

VOLUME 2

Operation Manual

Mazak VQC-20/40 SN 060231 - LinuxCNC/Mesa Retrofit

REVISION B - CONSOLIDATED FROM LIVE AUTHORITY

Field	Value
Document	MZK-LCNC-002
Revision	B
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Source commit	a3730f9
Machine	Mazak VQC-20/40, serial 060231
Original control	Mazatrol M-2 / MELDAS
Target control	LinuxCNC 2.9.10, Debian 13 PREEMPT_RT

STATUS - NOT READY FOR POWER OR MOTION

The repository is still at planning/bring-up stage. No live-power hold point is signed. Authority rows marked PROPOSED, COMMISSIONING_PENDING, HOLD_CONFLICT, or UNBOUND are not as-built evidence. The hardwired E-stop chain, resolver suffixes/scales, bitfile, analog polarity/scaling, field normal states, shared-bus behavior, and spindle terminals remain to be proven.

This volume replaces Rev A. Rev A must not be used for wiring, commissioning, or service.

APPLICABILITY HOLD

This is a commissioning-stage operating framework, not authorization to run the machine. Normal operator procedures become valid only after the corresponding signed hold points, as-built wiring, and commissioned values are incorporated into a later revision.

1. Control station and current state

The planned interface is LinuxCNC Axis with an optional WHB04B-style USB pendant. Physical panel-button reuse is incomplete; the original Cycle Start input is deferred and the manual clamp input remains HAL-unbound. Do not assume an OEM button has retained its former function.

Both drive-output-permit and spindle-output-permit initialize FALSE. ATC DRY_RUN is 1. These deliberate holds are configuration controls and must be changed only by reviewed edits tied to signed commissioning steps.

2. Power-up framework

- Confirm the day's signed hold point authorizes the intended energy level; stop if it does not.
- Inspect the machine envelope, Z support state, tool/spindle condition, fluid leaks, cabinet condition, and all E-stop devices.
- Verify the dedicated LinuxCNC NIC is up at the recorded address and no unreviewed network or BIOS change occurred.
- Apply control power only. Confirm both commissioning holds are FALSE and all physical outputs remain safe.
- Review LinuxCNC messages, packet-error, watchdog, resolver error, drive-fault, E-stop monitor, and field-input states before any higher hold point.
- Use only the reviewed commissioning procedure for DC bus, S-ON, brake, spindle, or motion. Never bypass a linked output with ad-hoc halcmd writes.

3. Homing

The current INI is configured for switch-based homing in the order Z (HOME_SEQUENCE 0), X (1), Y (2). This is a skeleton configuration, not proof that Z is safe to move first. The resolver emulated index is not accepted for axis homing.

Before any homing attempt, verify raw/complement selection for all six limits and three home switches, prove the hardware overtravel path, measure travel and stop margin, and complete the Z-brake sequence if Z will move. Change homing order if the signed hazard review selects a safer sequence.

Axis	Configured limits (in)	Max vel / accel	Home sequence	Status
X / Joint 0	-39.4094 to 0.0394	1.0 / 5.0	1	PLACEHOLDER
Y / Joint 1	-30.0394 to 0.0394	1.0 / 5.0	2	PLACEHOLDER
Z / Joint 2	-18.1496 to 0.0394	1.0 / 5.0	0	PLACEHOLDER

4. Jogging and first motion

- Normal jogging is prohibited until D8 resolver proof, D9 first-move plan, D11 travel survey, D14 network qualification, and the relevant signed hold point are complete.
- The first move uses a normal closed-loop LinuxCNC jog, one physically isolated axis, at the recorded clamp and clearance. It does not use open-loop PID bias on an installed axis.
- Abort for wrong direction, unexplained zero-command motion, following error, resolver error, drive alarm, limit transition, packet error, watchdog, brake discrepancy, or loss of observer/instrument.

5. Spindle operation

The planned spindle path is 7i49 AOUT3 for analog speed and 7i84U-A OUT0/OUT1/OUT2 for FWD/REV/RUN. ORCM1 on OUT4 is a discrete orient command; it is not an analog position reference. Exact FR-SX model, terminals, polarities, analog mode, and READY handshake remain unverified.

Do not rotate the spindle until the spindle hold point is signed. Prove zero speed, at-speed, fault, run/direction exclusivity, analog scaling, gear confirmation, ORCM1-to-ORA1 timing, and cleanup after cancel/fault in both gear ranges.

6. Tool changer and magazine

Automatic M6 is blocked: [ATC] DRY_RUN remains 1, the Y soft-limit extension component is not implemented, the ATC hazard analysis/dry-cycle fixture is missing, and several solenoid identities/directions remain HOLD_CONFLICT.

A stalled or aborted ATC must not be jogged out under automatic power. Isolate energy, assess mechanical state, support any suspended tool, and use the approved manual recovery procedure after documenting the failure state.

ATC prerequisite	Current position
Spindle oriented	ORA1 planned on 7i84U-A IN4; terminal/timing unverified
Spindle zero speed	SZS planned on IN5; terminal/polarity unverified
Tool clamp / unclamp	PRS-9 IN15 and PRS-8 IN16; valve topology unverified
Magazine position	Five BCD inputs IN19-IN23, valid only with mag-in-pos IN28
Y/Z ATC zones	PRS-55 IN0 and PRS-66 IN1; fitment and dynamic-limit logic incomplete
Magazine direction	OUT13/OUT14 remain HOLD_CONFLICT pending SOL-8A/8B trace

7. G-code and tool data

- Use LinuxCNC G-code semantics, not Mazatrol conversational behavior, unless an explicit remap implements the function.
- Keep tool.tbl synchronized with verified physical pockets and the magazine BCD decode before any M6.
- Dry-run programs with spindle and drive outputs held off until coordinates, limits, units, work offsets, and remaps are independently checked.
- Parameter and program recovery from the M-2 is evidence for conversion/rollback; it is not automatically a LinuxCNC parameter import.

8. Coolant, air, hydraulics, and optional controls

Function	Authority	Restriction
Hydraulic/head-lube pump	7i84U-A OUT3	Proposed; prove pressure and contactor interface
Flood coolant pump	7i84U-A OUT11	Commissioning pending; interposing relay as required
Flood valve	7i84U-B OUT7	Proposed separate valve
Air / touch / tap blasts	7i84U-B OUT3/4/5	100 VAC legacy loads; RLY-5/6/7 required
Mist coolant	7i84U-A OUT12	HOLD_CONFLICT; do not energize
Magazine cover close	7i84U-B OUT8	Proposed single-coil function; trace before use
WHB04B pendant	USB / HAL	Optional; install only after base controls are proven

9. Shutdown and abnormal stop

- Use the reviewed normal stop sequence; wait for zero speed and confirmed stationary axes before machine-off.
- Confirm the Z brake is mechanically holding before torque is removed.
- Drop higher-energy hold points in the documented order; do not treat LinuxCNC exit as power isolation.
- After main power-off, retain lockout and measure the shared DC bus before service.
- After any E-stop, communications loss, drive alarm, resolver error, or ATC abort, require manual investigation and reset. Never auto-resume.

10. Operator alarm response

Indication	Immediate response
E-stop / hardware chain open	Keep clear; identify cause; do not reset until hardware state is known
Packet error or watchdog	Stop; preserve logs; inspect NIC/cable/bitfile; manual reset only after review
Resolver / following error	E-stop; isolate S-ON; preserve trace; do not retry by increasing limits
Drive fault	Isolate hazardous power; record amplifier indication; inspect shared READY propagation
Spindle orient timeout	Drop ORCM1/run permits; do not retry automatically; verify gear/zero-speed/OR A1
ATC abort	Isolate hydraulics/motion; document mechanism state; use approved manual recovery
Low air / hydraulic pressure	Inhibit ATC and clamp motion; correct supply/leak before reset